Hyun Yoon

Postdoctoral scholar, **University of California, Berkeley**

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Research Profile: I study how microbial and geochemical processes govern the fate and transformation of contaminants, integrating environmental microbiology, interfacial biogeochemistry, and predictive modeling.

EDUCATION

Ph.D. **Cornell University**, Ithaca, New York — 2019–2023
Civil and Environmental Engineering

M.S. **Korea Advanced Institute of Science and Technology (KAIST)**, Daejeon, South Korea — 2016–2018
Civil and Environmental Engineering

B.S. **Yonsei University**, Seoul, South Korea — 2010–2016
Atmospheric Sciences
- Exchange Program, ***University of California, Los Angeles*** *in 2014 (1 year)*

RESEARCH EXPERIENCE

Postdoctoral Scholar, **University of California, Berkeley** — 2024–present
& NIH Superfund Research Program Trainee
- Mentors: Profs. Lisa Alvarez-Cohen & David L. Sedlak
- *Research focus*: Environmental microbiology and geochemistry governing contaminant fate and transformation, with a primary focus on PFAS

Graduate Research Assistant, **Cornell University** — 2019–2023
- Advisor: Prof. Matthew C. Reid
- *Research focus*: Wetland biogeochemistry with a focus on arsenic and carbon dynamics

Graduate Research Assistant, **KAIST** — 2016–2018
- Advisor: Prof. Sukhwan Yoon
- *Research focus*: Biological treatment of N_2O emissions from wastewater treatment plants

HONORS AND AWARDS

National Interest Waiver (NIW), **United States Citizenship and Immigration Services** — 2025
(U.S. permanent residency granted based on publication record and expert references)

International Research Travel Grant, **Cornell University** — 2022
(Proposal selected, funding two-month visit to uOttawa for on-site collaboration)

Featured in Journal Spotlight, ***Applied and Environmental Microbiology*** — 2022
(Selected by the editorial team in the respective issue)

Cornell Fellowship, **Cornell University** — 2019–2020
(Competitively awarded full funding package for the first year of Ph.D. study)

KAIST Support Scholarship for State Scholarship Student, **KAIST** — 2016–2018
(Competitively awarded full funding package for the duration of the M.S. program)

Internal Scholarship & Academic Awards, **Yonsei University** — 2015
(Competitively awarded final semester tuition funding)

Excellence Award, **Republic of Korea Army** Squad Leader Training Course — 2012
(Awarded for ranking 2nd overall among ~200 trainees)

PUBLICATION RECORD

Peer-reviewed Journal Publications:

1. Sarah Choyke[†], **Hyun Yoon**[†], Caroline Rose Alukkal, Youn Jeong Choi, Collin Meyers, Jonathan Thorn, Hannah Kruelle, Anthony Grimaldi, Ben Priest, Taryn McKnight, Josie King, Ananya Visweswaran, James Gray, Linda S. Lee, Lisa Alvarez-Cohen, David L. Sedlak, Fuhar Dixit and Mohamed Ateia. 2026. Versatile TOP Assay Surrogates for Evaluating Oxidation Efficiency in Complex Environmental Samples. ***Journal of Hazardous Materials Letters***. 7. 100198. ***([†]Equal first author contribution)***
2. Fuhar Dixit, Sarah Suhnholz, **Hyun Yoon**, Elizabeth M. Fries, Edmund H. Antell, Christopher P. Higgins, Lisa Alvarez-Cohen and David L. Sedlak. 2026. Evaporative losses of perfluoroalkyl sulfonamides during PFAS analysis. ***ACS Omega***, 11(9), 14638-14643.
3. Elisa Garabello, **Hyun Yoon**, Matthew C. Reid and Andrea Giometto. 2026. Tunable low-rate genomic recombination with *cre-lox* in *e. coli*: a versatile tool for synthetic biology and environmental sensing. ***Applied and Environmental Microbiology***, 92(4), e1768-25.
4. **Hyun Yoon**, Fuhar Dixit, Jonathan J. Uhler, Anna-Ricarda Schittich, Edmund H. Antell, Lisa Alvarez-Cohen and David L. Sedlak. 2026. Effects of organic matter, iron (hydr)oxides, and iron reductive dissolution on PFAS sorption in AFFF-impacted solids. ***Environmental Science & Technology Letters***, 13(1), 130-136.
5. **Hyun Yoon**, Michael Vega, Jiaxing Wang, Alexandre J. Poulain, Andrea Giometto, Ludmila Arstilde and Matthew C. Reid. 2024. Repression of arsenite uptake and methylation by dissolved organic matter. ***Environmental Science & Technology Letters***, 11(8), 838-844.
6. Heejoo Han, Daehyun D. Kim, Min Joon Song, Taeho Yun, **Hyun Yoon**, Hong Woon Lee, Young Mo Kim, Michele Laureni and Sukhwan Yoon. 2023. Biotrickling filtration for the reduction of N_2O emitted during wastewater treatment: results from a long-term *in situ* pilot-scale testing. ***Environmental Science & Technology***, 57(9), 3883-3892.
7. **Hyun Yoon**, Benjamin Stenzler, Lena Abu-Ali, Maria P. Asta, Alexandre J. Poulain and Matthew C. Reid. 2023. Effects of iron and dissolved organic matter on bioavailability of arsenite under anaerobic conditions. ***ACS ES&T Water***, 3(11), 3676-3686.
8. **Hyun Yoon**, Andrea Giometto, Martin P. Pothier, Xuhui Zhang, Alexandre J. Poulain and Matthew C. Reid. 2022. Time-dependent biosensor fluorescence as a measure of bacterial arsenic uptake kinetics and its inhibition by dissolved organic matter. ***Applied and Environmental Microbiology***, 88(16), e00891-22. **Featured in *AEM* Research Spotlight**
9. Lena Abu-Ali, **Hyun Yoon** and Matthew C. Reid. 2022. Effects of organic sulfur and arsenite/dissolved organic matter ratios on arsenite complexation with dissolved organic matter. ***Chemosphere***, 302, 134770.
10. Daehyun D. Kim, Doyoung Park, **Hyun Yoon**, Taeho Yun, Min Joon Song and Sukhwan Yoon. 2020. Quantification of *nosZ* genes and transcripts in activated sludge microbiomes with novel group-specific qPCR methods validated with metagenomic analyses. ***Water Research***. 185, 116261.
11. Min Joon Song, **Hyun Yoon**, Tae Kwon Lee, Miye Kwon and Sukhwan Yoon. 2020. Orthophosphate enhances abiotic nitrous oxide production from aerobic hydroxylamine decomposition. ***Biology and Fertility of Soils***, 56(5), 687-695.
12. **Hyun Yoon**, Min Joon Song, Daehyun D. Kim, Fabrizio Sabba and Sukhwan Yoon. 2019. A serial biofiltration system for effective removal of low-concentration nitrous oxide in oxic gas streams: mathematical modeling of reactor performance and experimental validation. ***Environmental Science & Technology***, 53(4), 2063-2074.
13. **Hyun Yoon**, Min Joon Song and Sukhwan Yoon. 2017. Design and feasibility analysis of a self-sustaining biofiltration system for removal of low concentration N_2O emitted from wastewater treatment plants. ***Environmental Science & Technology***, 51(18), 10736-10745.

Selected Manuscripts under review and in preparation:

1. **Hyun Yoon**[*], John Han, Lisa Alvarez-Cohen and David L. Sedlak. Molecular structure and environmental context jointly predict PFAS biotransformation outcomes: a literature-derived machine learning analysis. *In prep.* ***([*]Correspondence)*** [Research Poster Link]

2. Anna-Ricarda Schittich[†], **Hyun Yoon**[†], Leyan Peng, Fuhar Dixit, Tae-Kyoung Kim, Fanshu Geng, Damian Helbling, Christopher Olivares, David L. Sedlak and Lisa Alvarez-Cohen. Charge-dependent sorption controls the bioavailability and biotransformation of PFAS precursors in AFFF-impacted solids. *In prep.* ***([†]Equal first author contribution)*** [Research Poster Link]
3. **Hyun Yoon**[†], Michael Vega[†] and Matthew C. Reid. Genome-resolved metatranscriptomic analysis of arsenic demethylation and detoxification in a methanogenic rice paddy soil. ***bioRxiv***, 2026-08 [Link] ***([†]Equal first author contribution)***
4. Sarah Suhnholz, Boris Droz, **Hyun Yoon**, Corey De La Cruz, Nayantara T. Joseph, Johanna Peeters Weem, Trevor Schwichtenberg, Eric Redman, Mark Benotti, Gerrad D. Jones, Damian E. Helbling, Lisa Alvarez-Cohen, David L. Sedlak, Jennifer A. Field, Christopher P. Higgins and Fuhar Dixit. PFAS forensics strategy for source identification: an intermethod comparison across environmental challenge matrices. *In prep.*

Technical Reports & Government Documents:

1. Fuhar Dixit, **Hyun Yoon**, Lisa Alvarez-Cohen and David L. Sedlak, 2026. A simple and robust forensic technique for differentiating PFAS associated with AFFF from other PFAS sources. ***SERDP Final Report***, ER20-1330. [Link]

GRANT ACTIVITIES

Project 3 Proposal Lead, **UC Berkeley Superfund Research Program (NIEHS P42)** (USD 3.8M proposed) — 2026 (*Under Review*)

- Title: "PFAS Release from Long-Term Sources and Their Fate in Subsurface Environments" (FY2027–2032 Renewal)
- Role: Spearheaded the proposal development of engineering portion (Project 3) for the 5-year center renewal. Authored technical progress reports of the previous term, synthesized preliminary fate/transformation data, and actively collaborated with biomedical teams (Projects 1 & 2) to integrate engineering outcomes with biological exposure pathways across campus.

Proposal Lead, **U.S. Department of Defense, SERDP** (USD 1.7M proposed) — 2025 (*Submitted/ not funded*)

- Title: "Effect of Subsurface Solids Properties and Redox Conditions on PFAS Transformation at AFFF-Impacted Sites"
- Role: Led the development of the pre-proposal and full proposal.

Postdoctoral Project Lead, **U.S. Department of Defense, SERDP** (USD 1.8M) — 2024–2027 (*Funded*)

- Title: "Assessment of Physical, Chemical, and Biological Factors Controlling Biotransformation of Cationic and Zwitterionic Precursors in PFAS Source Zones (ER23-3827)"
- Role: Led day-to-day research implementation and project coordination, including experimental planning, data collection and analysis, budget management, mentoring of team members, progress presentations and reporting, and coordination with project collaborators.

Postdoctoral Project Lead, **U.S. Department of Defense, SERDP** (USD 1.5M) — 2021–2025 (*Funded*)

- Title: "A Simple and Robust Forensic Technique for Differentiating PFAS Associated with AFFF from Other PFAS Sources (ER20-1330)"
- Role: Led day-to-day research activities and project coordination, including data collection and analysis, mentoring of team members, progress reporting, and collaboration across participating research groups.

Principal Investigator, **International Research Travel Grant, Cornell University** (USD 3K) — 2022 (*Funded*)

- Title: "Effects of Iron and Organic Matter on Arsenic Bioavailability Using Novel Anaerobic Biosensor Techniques"

- Role: Developed the full proposal and conducted a two-month research collaboration with Prof. Alexandre Poulain's group at the University of Ottawa.

Graduate Researcher / Project Contributor, **NSF Environmental Chemical Sciences Program, Division of Chemistry** (USD 355K) — 2019–2022 (*Funded*)
- Title: "Arsenic Complexation with Reduced Organosulfur Moieties in Soil Organic Matter: Implications for Arsenic Oxidation via Biotic and Abiotic Pathways"
- Role: Post-award team member, data collection and analysis, collaborative partnerships.

Graduate Researcher / Project Contributor, **USDA-NIFA Bioenergy, Natural Resources, and Environment Program** (USD 495K) — 2018–2021 (*Funded*)
- Title: "Closing the Si Cycle in Rice Agroecosystems to Sustainably Control As and Cd Uptake by Rice Grown Under Alternate Wetting and Drying (AWD)"
- Role: Post-award team member, data collection and analysis, field sampling campaign to Arkansas rice paddies, collaborative partnerships.

Graduate Researcher / Project Contributor, **Ministry of Environment of Korea** (USD 861K) — 2019–2022 (*Funded*)
- Title: "Mitigation and Control of Low-Level N_2O Emissions in Wastewater Treatment Plants (Global Top Environmental R&D)"
- Role: Contributed to the development of the full proposal, provision of seed data, lab-scale and pilot-scale biofilter operation, project management, training of members, collaborative partnerships.

TEACHING AND MENTORING EXPERIENCE

Teaching:

Guest lecturer, **Seoul National University** — May 2026
- Provided an invited lecture titled "Predicting PFAS Fate in Subsurface Environments via Biogeochemical Controls" and discussion section for course "*CEE 457.634: Contaminant Transport Analysis* (3 credits; 15 students)"

Teaching assistant, **Cornell University** — Spring 2022, Spring 2023
- Teaching assistantship for course "*CEE 4530: Laboratory Research in Environmental Engineering* (3 credits; 15-20 students)"
- Led lab sessions, office hours, and student project management
- Delivered hands-on instruction on analytical instrumentation and data analysis for undergraduate researchers.

Instrument manager of Biogeochemistry and Ecological Eng. Lab, **Cornell University** — 2021–2023
- Provision of regular training on the operation and maintenance of analytical instruments, including (HPLC-)ICP-MS, IC, and TOC analyzer

Mentoring:

Research advisor, **University of California, Berkeley** — 2024–present
- Mentoring and training on essential protocols around PFAS analysis to Ph.D. students (Ananya Visweswaran, Nicole Garrido, Leyan Peng) and undergraduate researchers (Olivia Wang, Peter Reyes, Diego Canela)

PEER-REVIEW ACTIVITIES

Ad-hoc Reviewer for:

- *Environmental Science & Technology* (16 reviews completed)
- Other peer-reviewed journals: *Applied and Environmental Microbiology, ACS Omega, Agrosystems, Geosciences & Environment*

INVITED TALKS

1. **Invited research seminar,** December 2026, *AEESP Future Faculty Seminar Series* | Host: Association of Environmental Engineering and Science Professors (AEESP) (scheduled)
2. **Invited lecture for course "*CEE 457.634: Contaminant Transport Analysis*",** May 2026, "Predicting PFAS Fate in Subsurface Environments via Hydrologic and Biogeochemical Controls," *Department of Civil and Environmental Engineering,* ***Seoul National University***, Seoul, South Korea | Host: Dr. Yongju Choi
3. **Invited research seminar,** April 2026, "Predicting PFAS Fate in Subsurface Environments via Hydrologic and Biogeochemical Controls," *Department of Civil and Environmental Engineering,* ***Korea Advanced Institute of Science and Technology (KAIST)***, Daejeon, South Korea | Host: Dr. Sukhwan Yoon
4. **Invited research seminar,** March 2026, "Redox and Soil–Water Chemistry Controls on Contaminant and Nutrient Dynamics," *Department of Biological and Agricultural Engineering,* ***Kansas State University***, Manhattan, KS | Host: Dr. Mark Wilkins
5. **Invited research seminar,** July 2025, "Regulation of Arsenic Methylation and Demethylation Dynamics by Organic Carbon in Rice Paddy Soil & Effects of Iron Minerals and Reductive Dissolution on PFAS Precursor Sorption in AFFF-Impacted Soils," *Department of Green Chemistry,* ***University of Chemistry and Technology, Prague (UCT Prague),*** Czech Republic | Host: Dr. Lenka McGachy

CONFERENCE PRESENTATION

1. **Hyun Yoon**, John Han, Lisa Alvarez-Cohen and David L. Sedlak. 2026. Machine Learning Reveals How Molecular Structure and Environmental Conditions Jointly Drive PFAS Biotransformation. ***Superfund Research Program Annual Meeting.*** Portland, OR, *poster*
2. **Hyun Yoon**, Anna-Ricarda Schittich, Fuhar Dixit, Jonathan J. Uhler, Leyan Peng, Fanshu Geng, Damian Helbling, Lisa Alvarez-Cohen and David L. Sedlak. 2026. Charge-Dependent Sorption Controls the Bioavailability and Biotransformation of PFAS Precursors in AFFF-Impacted Solids. ***Gordon Research Conference 2026 Environmental Sciences: Water.*** Plymouth, NH, *poster*
3. **Hyun Yoon**, Fuhar Dixit, Jonathan J. Uhler, Anna-Ricarda Schittich, Edmund H. Antell, Lisa Alvarez-Cohen and David L. Sedlak. 2025. How organic matter and iron mineral dynamics, including reductive dissolution, affect PFAS sorption to AFFF-contaminated solids. ***Berkeley Energy & Resources Collaborative (BERC) 2025 Resources Symposium: Water as a Resource***. Berkeley, CA, *poster*
4. Michael Vega, **Hyun Yoon** and Matthew C. Reid. 2025. Regulation of arsenic methylation and demethylation dynamics by organic carbon in rice paddy soil. ***Goldschmidt 2025 Conference***. Prague, Czech Republic, *oral*
5. **Hyun Yoon**, Anna-Ricarda Schittich, Leyan Peng, Nicole Garrido, Fuhar Dixit, Christopher Olivares, David Sedlak and Lisa Alvarez-Cohen, 2024. Assessment of physical, chemical, and biological factors controlling biotransformation of cationic and zwitterionic precursors in PFAS source zones. ***DOD Energy Innovation Symposium***. Washington, DC, *poster*
6. **Hyun Yoon**, Fuhar Dixit, Katerina Tsou, Yanghua Duan, David L. Sedlak and Lisa Alvarez-Cohen. 2024. Effects of iron, organic matter, and seawater intrusion on PFAS precursor sorption in soil and groundwater. ***SETAC North America 45th Annual Meeting***. Fort Worth, TX, *oral*
7. **Hyun Yoon**, Benjamin Stenzler, Lena Abu-Ali, Maria P. Asta, Alexandre J. Poulain and Matthew C. Reid. 2023. Using an anaerobic biosensor to model the effects of dissolved organic matter and complexed ferric iron on arsenite bioavailability. ***ACS Fall 2023 National Meeting***. San Francisco, CA. *oral*
8. **Hyun Yoon** and Matthew C. Reid. 2023. What is the role of carbon catabolite repression in regulating

microbial uptake and methylation of arsenic? ***2023 AEESP Research & Education Conference***, Boston, MA. *poster*

9. **Hyun Yoon**, Martin P. Pothier, Xuhui Zhang, Andrea Giometto, Alexandre J. Poulain and Matthew C. Reid. 2021. Kinetic study of microbial arsenite uptake using whole-cell biosensor: Inhibitory effects of thiols and low molecular weight organic molecules. ***ACS Spring 2021 National Meeting***, Virtual meeting. *oral*
10. **Hyun Yoon**, Min Joon Song, Daehyun D. Kim and Sukhwan Yoon. 2018. Serial biofiltration system for effective removal of low-concentration nitrous oxide in oxic gas streams. ***17th ISME***, Leipzig, Germany. *poster*
11. **Hyun Yoon** and Sukhwan Yoon. 2017. Removal of low concentration nitrous oxide from gas streams using a self-sustaining biofiltration system with gravitationally-fed wastewater as the nutrient solution. ***ASM Microbe 2017***, New Orleans, LA. *poster*

PATENT

1. Sukhwan Yoon, **Hyun Yoon**, Min Joon Song, Heejoo Han and Daehyun D. Kim. Apparatus for removing nitrous oxide by using biofilters. ***Korea Patent*** No. 10-2227395-0000, filed Apr 2019 and issued Mar 2021.
2. Sukhwan Yoon and **Hyun Yoon**. The self-sustaining biofiltration apparatus for removal of nitrous oxide and the method of its installation and operation at wastewater. ***Korea Patent*** No. 10-1923992-0000, filed Jan 2017 and issued Nov 2018.

LEADERSHIP & ACTIVITIES

Graduate Student Association of Civil and Environmental Engineering, **Cornell University** 2020–2021
- Role: Social media chair

Glee Club (Choir), **Yonsei University** 2013
- Role: Baritone, Club executive

Squad Leader as Tank Driver, **Republic of Korea Army** 2011–2012
- Rank at Discharge: Sergeant
- Led a squad comprising three tanks and 12 soldiers